

LinT_EX: Enhancing L^AT_EX document accessibility for screen reader users

by
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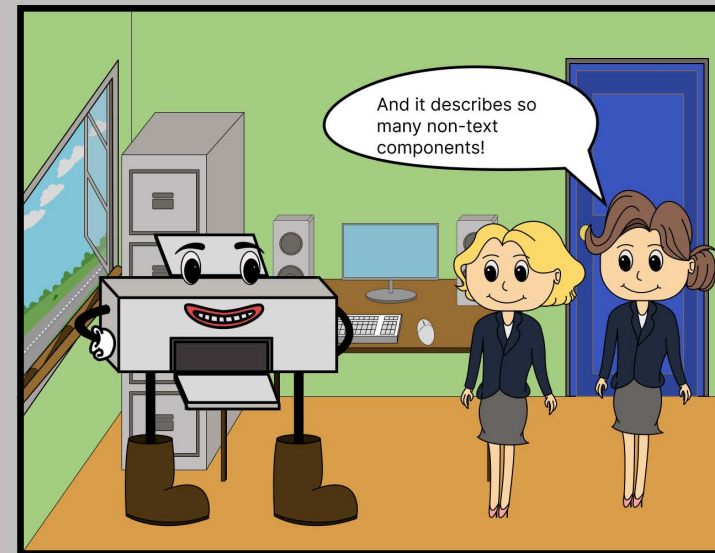
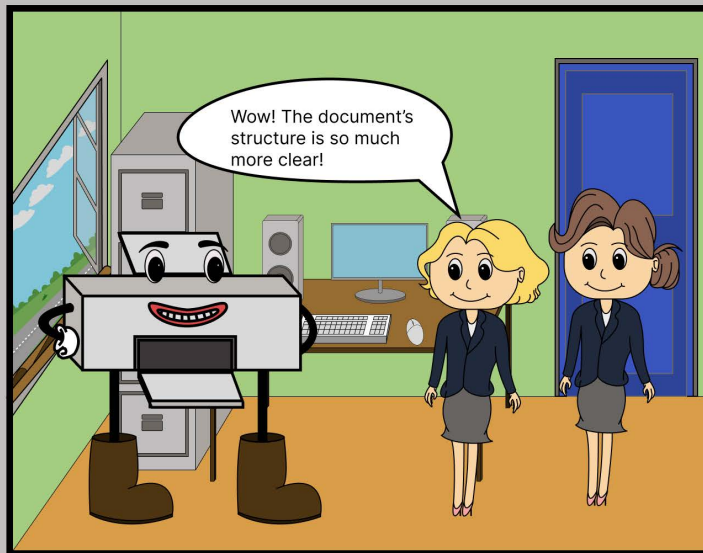
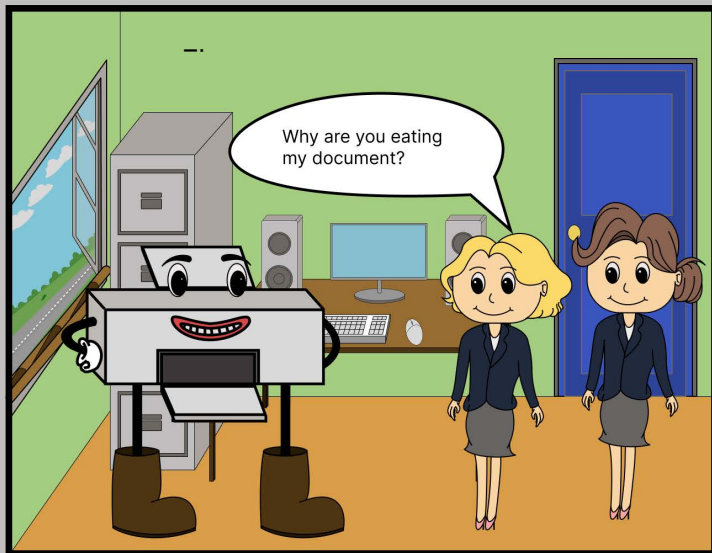
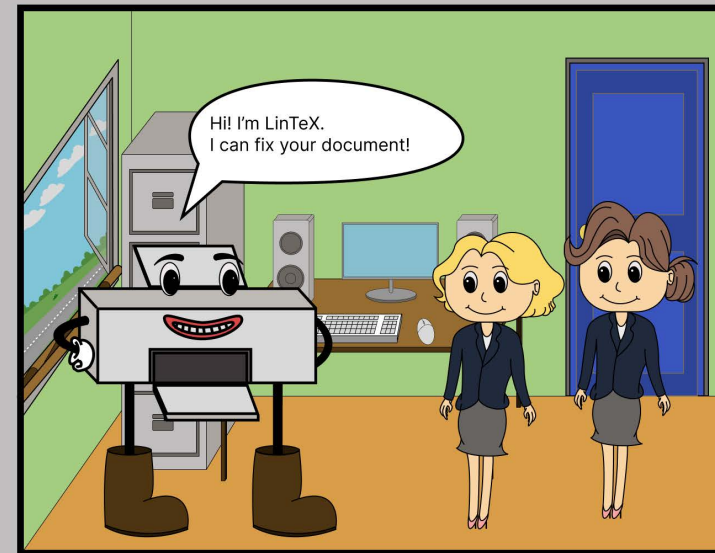
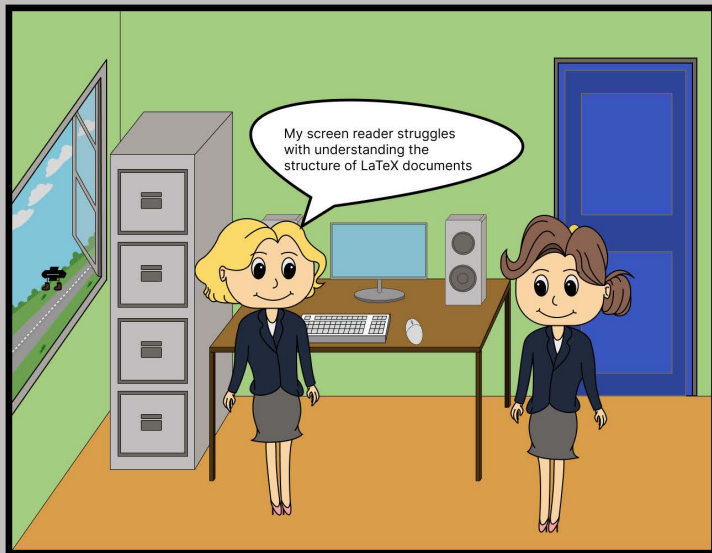
Department of
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Abstract

The abstract is a bird's eye view of the project.

It should not exceed one page. Mention the scope and objectives of the project, the methodology used, the main findings, and significance of your results.



Acknowledgements

I would like to thank my supervisor, Brian Mitchell, for his continuous guidance and support throughout the project. His depth of knowledge in related topics and the management of the project overall had vastly improved the project outcome. Although Brian would likely say luck is just a probability taken personally, I consider myself lucky to have come across and taken an interest in his project description.

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Glossary

accessible document	an accessible $\text{\LaTeX}$ document created by applying LinTeX to the inaccessible document. This document can be found in section A.7. iv, 8, 16, 17, 22, 24, 42
AI	artificial intelligence 1
alternative text	text used in place of non-text components for accessibility purposes. 2, 8
AST	abstract syntax tree 6
BDD	behaviour-driven development 8, 9
CDN	content delivery network 7
CGI	Common Gateway Interface 4
CLI	command-line interface 7
CMD	command prompt for Windows 7
CSS	Cascading Style Sheet 7
GUI	graphical user interface iii, 1, 11
HTML	Hypertext Markup Language 6, 7
HTTP	HyperText Transfer Protocol 7
IDE	integrated development environment 11, 26
inaccessible document	a $\text{\LaTeX}$ document that has intentionally been made inaccessible for screen reader users, which can be found in section A.6. iv, v, 11, 16, 17, 22, 24, 40
JS	JavaScript 7
likert scale datum	a scale, usually with 5 or 7 points, indicating a participant's agreement or disagreement with some topic (Sullivan 2013). 10, 11
metadata	data about data, such as a file's structure. 18, 22
npm	Node package manager 7
OS	operating system iv, 9
PDF	Portable Document Format 1, 6
PEG	parsing expression grammar iii, iv, 4–6, 8, 15, 44
qualitative data	data that cannot be represented numerically, such as descriptions, statements or quotes (National Library of Medicine 2026a). 10

quantitative data	data that can be represented in numerical form (National Library of Medicine 2026b). 10
right-tailed test	a test for whether the hypothesis should be rejected (Exeter 2025). 24
tagging	the process of providing data about the structure of a document to be used in its audio output. 1, 2, 26
TDD	test-driven development 8, 9
think-aloud session	users think aloud while performing a given task, with the goal of identifying areas of improvement in the usability of the given context. 10, 11
UI/UX	user interface and user experience 11
WSL	Window's Subsystem for Linux 6

1 Introduction

1.1 Identifying $\text{\LaTeX}$'s key accessibility problems for screen reader users

Similar to a tool like Microsoft Word, $\text{\LaTeX}$ is a tool used to process documents. $\text{\LaTeX}$ does this in a programmatical manner, likening it to a programming language, whereas Microsoft Word has a graphical user interface (GUI) which simplifies the process for an ordinary user. However, $\text{\LaTeX}$ prevails in its ability to process scientific documents which may require graphs, images, tables, or advanced mathematical capabilities.

$\text{\LaTeX}$ struggles to implement accessibility for screen readers, largely due to it not originally being built for that purpose, as explained in section 1.3. Additionally, $\text{\LaTeX}$ is broken by design, originally created as a set of macros built on the $\text{\TeX}$ typesetting engine that has evolved to the $\text{\Lua\LaTeX}$ that we can work with in modern times.

Additionally, Portable Document Format (PDF) files are built to not be changed, which adds difficulties in attempting to add tagging.

1.2 Why accessibility is important

Accessibility has the potential to assist all users, even if it is aimed a few focus groups, with one such case being where an environment, such as a noisy train, restricts the user from hearing the audio output of a video. In this context, an accessibility feature such as subtitles on a video, ideally without artificial intelligence (AI) generation, can greatly enhance a user's experience.

For example, in Northern Ireland alone, there are an estimated 8,000 users with sight loss (University 2026). These users may require the assistance of screen readers or someone else to read it out to them, which can add to their stresses, with the context simply not being built with their needs in mind.

Users with disabilities are over 50% more likely to face difficulties due to a lack of digital inclusion (University 2026), which highlights the severity of its impact on a website's overall traffic. These users should not be an afterthought of the website's design, but rather a key focus.

1.3 Why LuaLaTeX provides better accessibility than PDFLaTeX

PDF $\LaTeX$ was created to be a media printer, such that it produced PDF documents with the sole purpose of being printed out. Due to its innate design, this means it lacked support for screen reader users. PDF $\LaTeX$ also frequently has outdated, or frankly broken, workarounds to implement accessibility, such as the accessibility package (Clifton 2019) which has been deprecated since February 2021. Additionally, as its focus has shifted to maintenance rather than development, PDF $\LaTeX$ is unsuitable for accessibility where requirements change frequently, making it an outdated $\TeX$ engine, which unfortunately still sees frequent usage.

The introduction of Xe $\TeX$ improved font handling in $\LaTeX$, yet its lifetime was short-lived, with it almost being considered an evolutionary deadend, at the addition of Lua $\LaTeX$.

Unlike PDF $\LaTeX$, Lua $\LaTeX$ was created for the purpose of creating electronic documents, giving its development a suitable reason to support screen reader users through use of alternative text and tagging. Our tool, Lin $\TeX$, utilises Lua $\LaTeX$'s accessibility support, with emphasis placed on screen reader users.

1.4 Aims and objectives

This project aimed to enhance $\LaTeX$ documents' accessibility, with a focus placed on screen reader users, by creating a prototype, Lin $\TeX$, which analyses a $\LaTeX$ document to identify where to apply alternative text and tagging.

The following objectives were created for this context:

- OBJ1** The tool, Lin $\TeX$, should be maintainable, as accessibility often requires changes to its handling.
- OBJ2** Lin $\TeX$ should improve a document's audio output in terms of its contents, when read by a screen reader.
- OBJ3** Lin $\TeX$ should provide useful error messages and descriptive.
- OBJ4** Lin $\TeX$ should be reasonably accessible, following the accessibility guidelines defined in section 2.4.

1.5 Research questions

The following research questions cover the theoretical aspects of this project:

RQ1 What considerations are necessary to make a $\text{\LaTeX}$ document accessible for screen reader users?

RQ2 Why does $\text{\LaTeX}$ suffer a lack of accessibility for screen reader users?

2 Background Literature

2.1 Choice of programming language

There were many considerations when deciding what programming language to write LinTeX in. The languages chosen were based on those that I had prior experience in, to ensure minimal hindrance from the language itself to the development of the project. An important factor is ensuring LinTeX's maintainability in the future. Therefore, only the languages that have actively maintained parsing expression grammar (PEG) libraries will be considered, as can be seen in Table 2. The most recent update times provided in Table 2 are in reference to 24th November 2025.

Table 2: Languages and related PEG frameworks

Language	PEG library link	PEG library is in development
Python	https://pythonhosted.org/pyPEG2/	No, last updated in October 2015
Python	https://github.com/erikrose/parsimonious	Yes, last updated in November 2025
JavaScript	https://github.com/pegjs/pegjs	No, last updated in November 2019
JavaScript	https://peggyjs.org/	Yes, last updated in September 2025
Go	https://github.com/pointlander/peg	Yes, last updated in November 2025
Go	https://github.com/mna/pigeon	Yes, last updated in November 2025
C	N/A	N/A
Java	https://github.com/sirthias/parboiled	Yes, last updated in November 2025
C#	https://code.google.com/archive/p/peg-sharp/	No, last updated in May 2011
C#	https://github.com/otac0n/Pegasus	No, last updated in December 2023
C++	https://github.com/yhirose/cpp-peglib	Yes, last updated in July 2025
PHP	https://github.com/hafriedlander/php-peg	No, last updated in May 2014
TypeScript	https://github.com/metadevpro/ts-pegjs	No, last updated in November 2024
TypeScript	https://github.com/EoinDavey/tsPEG	Yes, last updated in November 2025
Dart	https://github.com/darmie/dart-peg	No, last updated in July 2015

Based on the current criteria, we can rule out C, C#, PHP, and Dart.

As many languages remain, there is a need for stricter criteria: suitability to be utilised as a web application as well as compatibility with the University of Strathclyde's development web server (devweb) system.

It should be noted that while Go, C, C#, and C++ are not directly compatible with devweb, as referenced in Table 3, they can be supported through the use of Common Gateway Interface (CGI) scripts, and consequently should not be fully disregarded for this alone. However, based on their support for web development, C, Java, C#, and C++ will have a lower priority for being a chosen language.

Table 3: Languages and related website development support

Language	Web development support	DevWeb support
Python	Yes	Yes
JavaScript	Yes	Yes
Go	Yes	No
C	No, except for rare exceptions	No
Java	Yes, but not generally used	Yes
C#	Yes, but not generally used	No
C++	Yes, but not generally used	No
PHP	Yes	Yes
TypeScript	Yes	Yes
Dart	Yes	Yes

This leaves us with a focus placed on JavaScript, Go, and TypeScript. However, JavaScript will be removed from consideration as a strictly typed language is preferred for our context. Therefore, leaving Go and TypeScript as the main considerations for this project. Finally, as I have more experience and a stronger preference towards Go as opposed to TypeScript, it was the chosen programming language for the project.

2.2 Choosing a PEG library under the Go programming language

Go has two libraries that are used in relation to PEG systems: Pigeon by Martin Angers (mna) and PEG by Andrew Snodgrass (pointlander). Direct comparisons with core components of both have been made in Table 4.

While examples are relatively similar, the simplicity of installation for Pigeon as opposed to PEG, combined with the depths of documentation both in generated code and outwith it, makes Pigeon a more ideal option. Additionally, its greater number of features gives $\text{LinT}_{\text{E}}\text{X}$ more to make use of. Therefore, Pigeon was the chosen library for the project.

2.3 Choice of frameworks and libraries

2.3.1 Backend frameworks and libraries

There were three options for web frameworks and packages, namely Gin (Martínez-Almeida 2014), Gorilla (The Gorilla Authors 2023), and net/http (The Go Authors 2009c). For $\text{LinT}_{\text{E}}\text{X}$, net/http was chosen for the key reason that it had the least barriers to entry, as there was a provided guide on the Go development website that was utilised in the

Table 4: Comparison between PEG and Pigeon

	Pigeon	PEG
Installation	Requires running a <code>go install</code> command and setting up environment variables.	Requires supporting bash scripting (through Window's Subsystem for Linux (WSL) for Windows users), as well as running a <code>go install</code> command, cloning the repository, and running <code>go generate && go build</code> .
Documentation	Contains a Godoc page, wiki page, an examples subdirectory, and a detailed README.md providing what PEG features are supported, command line interface usage, installation, and examples.	Contains a detailed README.md and a PEG file syntax file containing what PEG features are supported, installation steps, and example projects links.
Generated Code	Generated Go code can be used to configure settings on the parser, view statistics or error information, debug, enable or disable memoization, and parse readers, files, and strings. Additionally, the generated code is well documented, with several comments surrounding usage of public functions.	Generated Go code can be used to run the PEG script on a provided string, debug and output the abstract syntax tree (AST), output errors in parsing, and enable or disable memoization.
Examples	Provides three direct examples: detecting indentation in files, parsing JSON files, and a calculator example. Examples were kept short (regarding PEG scripts) with the calculator and indentation examples showcasing usage via a main function.	Provides two project links for examples on PEG usage: a natural date/time parser and a scientific names parser. Both projects have large PEG scripts, with most lines being simple to read. The natural date/time project was intuitive to use and understand program flow.

project setup. However, in hindsight, more consideration towards what web framework to use would have had a great impact on the project as a whole, as `net/http` is a lower level web framework than `Gin` or `Gorilla`, consequently requiring more management on `LinTeX`'s end. Some examples of difficulties that occurred are a lack of POST requests being easily supported and necessitating that server files are explicitly stated as opposed to implicitly defined within a given context.

The `html/template` (The Go Authors 2009b) package was included in the project. This package provided the ability to load Hypertext Markup Language (HTML) files through `Go`, while allowing information to be provided to them. The alternative choice was to provide a raw print output to the client, which made the package a far clearer winner.

For the project to be able to compile $\text{\LaTeX}$ documents to convert them into PDFs, it was necessary to either use a pre-built package or to compile $\text{\LaTeX}$ documents locally. Pre-built packages that were found include `go-latex` (T. I. Authors 2020) and `latex-fast-compile` (Kpym 2020). It was necessary to rule out `go-latex` as it emulated a $\text{\LaTeX}$ compiler, rather than directly linking to one. Additionally, it lacked documentation and necessary features,

making it unsuitable. Latex-fast-compile was also unsuitable due to using PDF $\LaTeX$ rather than Lua $\LaTeX$, which was desired.

Consequently, it became necessary to compile $\LaTeX$ documents directly. This was done through the use of command prompt for Windows (CMD), which allowed the 'lualatex' command to be ran, consequently compiling the requested file. It did, however, prerequisite that the hosting server had a $\LaTeX$ compiler in their backend, for the relevant Lua $\LaTeX$ engine. The os/exec (The Go Authors 2009a) package allowed for a CMD shell to be created within Go code, consequently allowing the 'lualatex' command to be run. Hence, compiling the $\LaTeX$ document that the user provides.

2.3.2 Frontend frameworks and libraries

As plain Cascading Style Sheet (CSS) is difficult to work with and maintain, it was necessary to decide on a CSS framework for the project. My prior experience in Bootstrap (T. B. Authors 2011) made it a close choice, but the opportunity to learn to utilise Tailwind CSS (Tailwind Labs 2017) was more enticing and eventually won the decision.

To implement Tailwind CSS, it was necessary to install its command-line interface (CLI) through Node package manager (npm). This permitted usage of Tailwind CSS without necessitating JavaScript (JS) within the codebase, and avoided workarounds such as a content delivery network (CDN). Additionally, due to difficulties in using Tailwind CSS, it was necessary to utilise a component library built on top of it, for which Flowbite (Inc. 2023) was chosen as it could be used as almost plug-and-play.

To ensure appropriate responsiveness of Lin $\TeX$, it was necessary to be able to introduce updates in real time. For this, HTMX (Software 2020) was utilised, as it is a tool that permits HTML tags other than the 'a' and 'form' tags to send HyperText Transfer Protocol (HTTP) requests to the server. For example, it is possible to use HTMX to create a HTTP request when a given key is pressed through its event management.

2.4 Guidelines for accessibility

The following key points have been used in the design of Lin $\TeX$ to ensure its accessibility, following WCAG 2.2 (W3C 2018) standards.

- The website should have all non-text content contain an alternative text (WCAG 2.2: Guideline 1.1).
- The website may be operable through the keyboard alone (WCAG 2.2: Guideline 2.1).
- LinT_EX should provide simple error messages.
- The website should allow for content to be distinguishable, including supporting resizing content and minimal colour usage (WCAG 2.2: Guideline 1.4).
- The website should be presentable in a simpler format if required (WCAG 2.2: Guideline 1.3).
- The website should include navigation features (WCAG 2.2: Guideline 2.4).
- The website should be predictable (WCAG 2.2: Guideline 3.2).
- The produced document by LinT_EX should contain appropriate tagging.
- The produced accessible document by LinT_EX should contain appropriate alternative text, actual text, or artifact.

2.5 Choosing a software development methodology

The available options under consideration for the project are the waterfall methodology, agile development methodology, test-driven development (TDD), and behaviour-driven development (BDD).

The waterfall methodology requires significant research early in the project to have success. However, as I lack experience in PEG script usage and many related topics for this project, it was likely that changes would be needed dynamically.

TDD was the next consideration, with a personal strong bias towards it due to its usefulness in being self-documenting and saving the hassle of writing the unit tests after creating the software, which can be a more draining process. However, as Pigeon, the PEG library chosen in choosing a PEG library under the Go programming language (section 2.2), generates Go files without a way to test individual components, it was not possible to create unit tests without it using complete L^AT_EX code for the feature that should be tested. Additionally,

creating unit tests can be time-consuming, which may be unsuitable for the short timeframe of the project.

BDD was removed from consideration for similar reasons as TDD, with the addition of it being more suitable for larger contexts such as games development, where it is a lot less practical to test every feature individually.

The final methodology, agile development methodology, allowed for code to be created quickly through its sprints. Additionally, as there was no prerequisite to create unit tests before creating the features, it saved a lot of time in development for the project. Consequently, agile development methodology was chosen for this project.

2.6 Screen reader choice

Through a small number of criteria, as showcased in Table 5, it was possible to reduce the choices for what screen reader to use in our context.

Table 5: Screen reader pricing status and OSes

Screen reader	Paid service	operating system support
JAWS	Yes	Microsoft Windows 10, 11, and server editions
Apple VoiceOver	No	Apple devices including iPhones, iPads, Macs, and AppleWatches
Windows Narrator	No	Windows 10 and 11
TalkBack	No	Android 6.0 Marshmallow onwards
ChromeVox	No	Primarily ChromeOS, with older versions supporting devices using Chrome browser
Supernova	Yes	Microsoft Windows 10, 11, server editions, and virtual environments
NVDA	No	Microsoft Windows 8.1 to 11

As a student, I do not have the leeway to purchase subscriptions for several screen readers, making those that require payments ruled out, even if it would just be in the form of a free trial. Additionally, there is an ethical dilemma in requiring payments for a necessary accessibility tool, which pushed me away from choosing them. Another criterion was that the Windows operating system (OS) should be supported, with emphasis placed on Windows 10 and Windows 11. These were chosen as they are the most common PC OS used, consequently targeting a larger base of users who may require screen readers.

Considering these criteria, NVDA and Microsoft Narrator were the final choices. When testing both on a placeholder document, emulating the experience of a screen reader user working with $\text{\LaTeX}$, I found that there was more difficulty with Windows Narrator in terms of having it read the correct extract on the document, often requiring using a mouse to get

to the correct location. Inversely, NVDA read the correct extract in full detail. Hence, by process of elimination to a degree, NVDA was the chosen screen reader for LinT_EX.

Adobe Acrobat Reader was used to work alongside NVDA to make the documents easier to navigate in more complex contexts, such as where the document may not be read as simply as top-to-bottom, left-to-right.

2.7 User study design choices

2.7.1 User study structure design

A combination of a think-aloud session and questionnaires was utilised in the creation of the user study.

A think-aloud session was performed where users think aloud while performing a given task, with the goal of identifying areas of improvement in the usability of the given context. This was utilised to acquire qualitative data regarding the usability of LinT_EX, with focus placed on most affected areas in Table 9.

It was deemed necessary to compare a L^AT_EX document before and after applying LinT_EX's changes to determine whether LinT_EX had a measurable impact on the screen reader accessibility of the documents. This comparison was created through a listening of the document's screen reader output at a given stage, and a questioning stage immediately after the relevant listening.

Additionally, it was possible that the participant's experience level in related topics would have an impact on their answers to questions or experience during the think-aloud session, consequently necessitating a small questionnaire to gauge experience levels.

2.7.2 User study question design

Factors that may influence a user's experience in the user study include their experience level in L^AT_EX, using screen readers, and listening to audio recordings of speaking. Likert scale data were used to gauge these experience levels in a numerical method, providing quantitative data.

Qualitative data were obtained from open-ended questions, with a clear focus placed on

each question, such that it would minimise confusion.

To gauge an understanding of how truly impactful an inaccessible document's impact is on a screen reader user's experience, participants will be questioned on what they could recall from the listening. However, as the participant will have worked with the document directly during the think-aloud session, it is redundant to include it in the final questioning.

It is necessary to understand what components of the relevant listening were found to be simple or difficult. However, these should be asked separately to ensure no confusion about what is being asked. The participant will be asked to provide their opinion on how well the screen reader audio output could be understood in terms of its contents, with a likert scale datum to provide quantitative data that can be analysed.

Participants were asked if they were likely to accommodate screen reader users in the future, to gain an understanding of whether LinT_EX had made doing so appear to be an easier process.

2.8 GUI design considerations

LinT_EX took strong inspiration from integrated development environments (IDEs) with regards to its user interface and user experience (UI/UX) design. This is because L^AT_EXin structure is similar to a programming language, even being Turing complete (Murrish 2012). Hence, an environment suitable for programming is suitable program L^AT_EXin, with some special considerations put aside for its quirks. For example, depending on the user's screen size and preferences, they may wish to have the output of the L^AT_EX documents taking up half of the screen, a separate browser window or monitor, or out of the way until compilation.

Additionally, online code editors such as LeetCode (Tang 2015), HackerRank (Ravisankar **and** Karunanidhi 2011), and CodeWars (Doctor **and** Hoffner 2012) were similar in nature to what LinT_EX would like to achieve regarding its UI/UX, and were consequently used as inspiration.

3 Specification & Design

- software development methodology usage + other stuff
- key engineering requirements
- GUI design
- design patterns used
- maintainability via documentation (readme) + a lot of comments

Describe all details of the design and procedures used to achieve the project objectives. Do this chronologically.

It should be detailed enough to allow for an assessment of the rigour of your process, and, in the case of research projects, in terms of how well grounded your research is in the research literature. In these cases, refer back to relevant sections in the previous chapter.

Say which software lifecycle approach you used e.g., Waterfall, Spiral, Agile.

How did you gather user requirements?

3.1 Methodology

Which methodology did you choose?

3.2 Analysis

How did you decide on the particular software artifact you decided to develop?

3.3 Requirements

Here you explain what the functional and non-functional requirements are. Explain how you prioritised them. See <https://www.nuclino.com/articles/functional-requirements> for more information.

Functional requirement: "The system must *do* [requirement]."

Non-functional requirement: "The system shall *be* [requirement]."

Well-written functional requirements typically have the following characteristics:

Necessary: Although functional requirements may have different priority, every one of them needs to relate to a particular business goal or user requirement.

Concise: Use simple and easy-to-understand language without any unnecessary jargon to prevent confusion or misinterpretations.

Attainable: All requirements you include need to be realistic within the time and budget constraints set in the business requirements document.

Granular: Do not try to combine many requirements within one. The more precise and granular your requirements are, the easier it is to manage them.

Consistent: Make sure the requirements do not contradict each other and use consistent terminology.

Verifiable: It should be possible to determine whether the requirement has been met at the end of the project.

3.3.1 Functional Requirements

This is the **WHAT** of your artifact.

Functional requirements are product features that developers must implement to enable the users to achieve their goals. They define the basic system behavior under specific conditions.

Functional requirements need to be clear, simple, and unambiguous. Examples:

- The system must send a confirmation email whenever an order is placed.
- The system must allow blog visitors to sign up for the newsletter by leaving their email.
- The system must allow users to verify their accounts using their phone number.

3.3.2 Non-Functional Requirements

This is the **HOW** of your artifact. Example non-functional requirement: “When the submit button is pressed, the confirmation screen must load within 2 seconds.”

3.4 Design

3.4.1 Interface Design

Explain how you used wireframes, and how you tested these to design the user interface.

3.4.2 System Design

Show how you designed your database (if appropriate) and how you designed your system architecture, and the individual parts. Use UML and an Entity Relationship diagram

4 Implementation

4.1 Technologies used

Parsing was complete using a parsing expression grammar (PEG) script, which generated a Go file through Pigeon.

4.2 website with server-client relationship

4.3 PEG Parsing design

- something on top-down approach, maybe specific details

The PEG script can be found in section A.8.

4.4 verification & validation

- (manual testing and error messages for server-side errors that should not occur in most cases)

5 User study

A user study is an approach taken to understand the requirements of users in relation to your application (Center 2024). It can be used to better understand user requirements for those who will be using the system and consequently improve design decisions.

This user study was given ethical approval, as can be viewed in ethical approval (section A.1).

5.1 Study design

The user study was designed with the goal of evaluating the effectiveness of the tool, LinTeX, and identifying areas of improvement for the tool. To meet these goals, the user study was split into six key stages:

1. An introductory stage where the goal is to identify where the participant's knowledge lies in related topics.
2. An initial listening of the inaccessible document, which was produced for this user study.
3. A questioning stage based on the initial listening of the inaccessible document.
4. A think-aloud session for understanding the accessibility of LinTeX.
5. A final listening of the newly created accessible document by using LinTeX.
6. A questioning stage based on the final listening of the accessible document.

These stages allowed for a comparison of the before and after of using LinTeX, through stages 2 and 5, as well as associated questioning stages 3 and 6, allowing both qualitative and quantitative data to be obtained. Additionally, the think-aloud session provides an opportunity to judge the usability of LinTeX.

Finer details for the user study can be found in the user study script (section A.5), with the reasoning behind their choices located in the user study design choices (section 2.7).

5.2 Hypotheses

There were several hypotheses that this user study aimed to prove or disprove:

H1 The accessible document created by LinTeX showed an improvement from the inaccessible document initially used in terms of the document's accessibility for screen reader users. This hypothesis is based on the objective ??.

H2 LinTeX is reasonably accessible. This hypothesis is based on the objective ??.

H3 LinTeX provided useful error messages. This hypothesis is based on the objective ??.

H4 The inaccessible document used in the initial listening is not memorable.

5.3 Results

The sample size for this user study was small, with 5 participants being recruited for it.

5.3.1 Memorable document extracts

The following table, namely Table 6, showcases the results from the question in the initial questioning (stage 3), which requests participants to summarise the document.

Table 6: Memorable aspects of document extracts

Document extract	Comment
Improving spoken skills	The majority of participants recalled spoken skills being improved through tongue twisters, yet there was minimal mentions of other techniques referenced in the inaccessible document.
House sale	A little over half of the participants had recalled a house being mentioned, with the currency and specific pricing often incorrect.
Favourite pet	All users recalled the extract about the cat being the favourite pet.
Muhammad and Noah name	All users recalled this extract, with finer details such as: whose name was more popular, the positions of the names, and what names were in the context were often incorrect.
comparisons in Scotland	All users recalled this extract, with finer details such as: the friends' names, John's name, the class codes, whether the extract referenced John's friends or general students, who was assigned what grades, and the grades themselves, were often incorrect.
Grade comparisons	

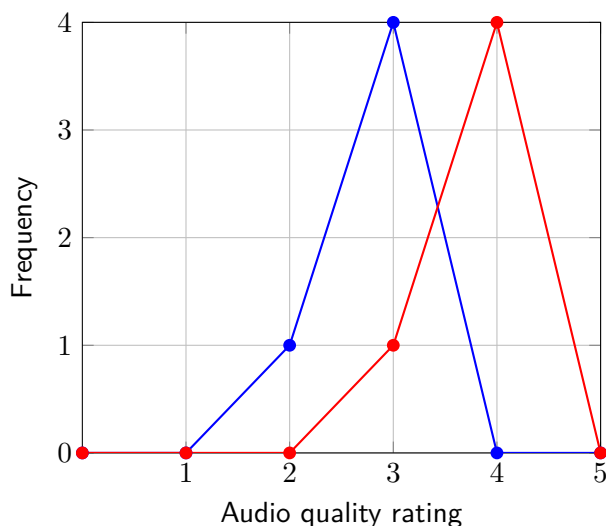
It should be noted that participants who had requested to pause or repeat a segment of the initial listening showcased a better recollection of the document extracts, with the average participant able to recall 4 extracts on average.

5.3.2 Accessible document and inaccessible document comparisons

The following figure, namely Figure 1, showcases the comparison of Likert scale data obtained from the user study that queries the participants about the quality of the screen

reader's audio output regarding its contents. This data was used to create a frequency polygon graph, for which the blue line showcases the data from the initial questioning (stage 3) and the red line showcases the data from the final questioning (stage 6).

Figure 1: Comparing screen reader output quality ratings before and after applying LinTeX



The key difficult components of both the initial listening (stage 2) and final listening (stage 5) can be found in Table 7. It should be noted that this is not an exhaustive list of all difficult components from both stages, but rather the key points that a majority of participants in the user study emphasised. For example, a few participants noted that the scale of numbers included in a table had an impact on its legibility.

Table 7: Difficult aspects from the initial and final listenings

Initial document	Final document
Repetition of 'percent' in the second table was found to be distracting. Images were difficult to keep track of when and where they had occurred. Images being plain file names made it difficult to ascertain their purpose.	Repetition of 'percent' in the second table was found to be distracting. Tables contained too much metadata, with focus placed on 'column N' being repeated for each table cell. Document metadata could be confused with the contents of the document, with emphasis placed on column numbers being mixed with name positions in the first table.
Headings and pages did not have sufficiently long pauses before beginning. Tables were read too fast with no pauses between table cells. The table containing names is viewed from top to bottom, whereas the screen reader reads from left to right.	

The key simple components of both the initial listening (stage 2) and final listening (stage 5)

can be found in Table 8. It should be noted that this is not an exhaustive list of all simple components from both stages, but rather the key points that a majority of participants in the user study emphasised. For example, many participants enjoyed the inclusion of an example tongue twister in place of the relevant image.

Table 8: Simple aspects from the initial and final listenings

Initial document	Final document
Standalone sentences were read slower and with pauses. Sentences separated by images or tables were easier to distinguish.	Standalone sentences were read slower and with pauses. Sentences separated by images or tables were easier to distinguish.
Lists provided in bullet point form were simple to understand due to a clear starting point.	Lists provided in bullet point form were simple to understand due to a clear starting point. Each point in the list started with 'bullet' instead of 'bullet point' which was the case previously, consequently speeding up the screen reader's output.
The table discussing grades is read left to right, whereas the table discussing names is read top to bottom. When the table can be viewed left to right, it followed the same order the screen reader used when creating the audio output.	The table discussing grades is read left to right, whereas the table discussing names is read top to bottom. When the table can be viewed left to right, it followed the same order the screen reader used when creating the audio output.
Supplementary sentences included before or after images and tables improved the overall clarity of the related extracts.	Supplementary sentences included before or after images and tables improved the overall clarity of the related extracts.
	Images were easier to understand due to their included descriptors.
	The screen reader included more pauses and spoke slower, with emphasis placed on tables.
	Structural indicators such as 'out of list', 'out of table', and 'table with M columns and N rows' improved tracking of where the screen reader was on the document at that moment.
	Reading the table heading before each table data cell improved tracking of what data correlated to what name in the first table.

It should be noted that all participants required less pausing or repetition of an extract in the final listening (stage 5) compared to the initial listening (stage 2). Additionally, another point of note is that tables were a frequent aspect that required pausing or repetition.

5.3.3 Applying student's paired t-test

The student's paired t-test was developed by William Sealy Gosset in 1908 (Wadhwa and Marappa-Ganeshan 2023). Its purpose, in our context, is to determine if the difference between our audio output quality Likert scale data in Figure 1 is significant.

Figure 2 has a standard deviation (s_d) of 0, as the difference between all results was 1. As a result of this, it leads to a division by zero. Depending on interpretation, this could be

Figure 2: Paired t-test formula

$$t = \frac{\bar{d} - \mu_0}{\frac{s_d}{\sqrt{n}}} \quad (1)$$

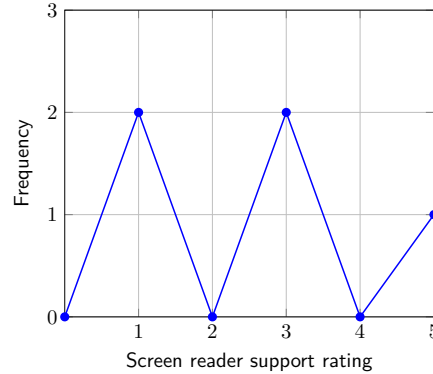
seen as t approaching infinity (∞) or simply an undefined result.

Under the basis of t approaching infinity, we can consequently view it as a large t value, which suggests there is statistical significance in the results.

5.3.4 Likelihood to support screen reader users

The following figure, namely Figure 3, showcases the results from the Likert scale data obtained in the final questioning (stage 6), which queries participants on their likelihood to support screen reader users in the future.

Figure 3: Participants likelihood to support screen reader users in the future



It should be put under consideration that all participants of the user study had indicated a lack of prior experience with screen readers.

5.3.5 LinTeX usability focuses

The following table, namely Table 9, highlights areas where usability could be improved or was found largely useful during the think-aloud session (stage 4). It should be noted that some results that stemmed from a lack of LaTeX knowledge were not included, as Table 9 aimed to identify the key components of usability.

Table 9: LinT_EX usability

Affected area	Comment
Uploading a file	Lacked responsiveness regarding actions performed when the upload button was clicked without a selected file, and informing the participant that a file has been uploaded.
Button phrasing	Phrasing of 'generate output' button caused participants to assume it could only be clicked after all errors were fixed.
Scrolling options	Scrolling options for the in-built code editor, errors, and the website as a whole, made navigation more difficult.
Submitting a document	A keybind to generate the document's PDF output was requested.
Error extracts	Most error extracts, the code taken from where the error had occurred, were helpful. However, a few syntax errors, such as a missing closing curly brace only provided the opening curly brace within its extract. It should be noted that many users utilised the browser's search features to locate where the error had occurred.
Image error messages	No difficulties found.
Table error messages	Attempts were made to place the 'tabular' group within the required argument of '\tagpdfsetup'.
Tagging error messages	Uncertainty existed with what should be provided to the required argument of '\DocumentMetadata' and how to handle several key-value pairs being provided to it, with some attempts to add an extra required argument.
Error messages	Examples helped to achieve the correct output, with long error messages found from requesting more details alleviating many prior concerns.
Text size	Text size was small.
Colour scheme	The colour scheme of the website may be inaccessible for users with dyslexia.

5.4 Analysis

5.4.1 Reasoning behind document extracts being memorable

This section uses data from Table 6 to make an analysis of the reasoning behind why specific sections were memorable, and others were not.

While the tongue twister was memorable in the initial listening (stage 2), likely due to it having a relatively large extract focused on it, participants had noted that providing the actual tongue twister made for a more enjoyable and consequently memorable experience in the final listening (stage 5).

Participants frequently misremembered finer details in smaller extracts relating to images, with only the extract on the cat being the favourite pet being the most memorable, likely due to its simplicity. This could be biased by the participants not being informed what specific questions would be asked, or the question being asked after the full document has been read, consequently requiring the participants to keep the full document in mind by that point. The latter point highlights a struggle a screen reader user may face.

Regarding name positioning and grade comparisons, a reasonable assumption as to why the overall topics were memorable despite finer details being frequently incorrect is that they required a greater degree of focus due to having more data in the tables. However, due to

the magnitude of the data, and the speed and lack of pauses in their listening, as referenced in Table 7, the particulars of the data were often mistaken. Additionally, supplementary sentences surrounding these tables often provided a general overview of what the tables had contained, which may have influenced a participant's understanding.

5.4.2 LinT_EX's impact on document screen reader accessibility

Figure 1 showcases that the accessible document's audio output rating had a higher peak compared to the inaccessible document. Additionally, we can view that the accessible document is shifted one data point higher on the x-axis at all points, with the same overall shape maintained as the inaccessible document. We can consequently conclude that, based on this quantitative data, LinT_EX had provided an improvement to the document's accessibility, with emphasis placed on screen readers.

Table 7 highlights the components of the initial listening (stage 2) and final listening (stage 5) that were difficult to understand. This permits a comparison between the two columns, for which we can determine that the majority of difficult aspects had been improved, as they were no longer in the final listening (stage 5)'s column. However, there remained two components that provided some confusion in the final listening (stage 5):

- The document's contents added unnecessary confusion at times, where it could be confused for metadata that the screen reader was reading out.
- The tables within the document contained too much metadata. The phrasing 'less is more' could be applied here, where too much information obstructed a participant's ability to understand its contents.

Table 8 showcases the fact that there were many more simple to understand components in the accessible document compared to the inaccessible document, with many of the previously helpful components remaining the same or having mild improvements. Consequently, we can deduce that simple components were retained with the same, or arguably better, quality.

5.4.3 Areas of improvement for LinT_EX usability

Table 9 highlights many areas where LinT_EX suffers in its usability. Hence, some proposed solutions for these concerns have been indicated below:

- The file upload process could be simplified by combining both the file selector and the upload file button. This would reduce the number of steps required on the user's part to upload the file and begin using LinT_EX.
- The phrasing of specific website components could be changed. For example, 'generate output' could be swapped out for 'check document' which has different connotations of applying some form of validation to the document. Additionally, a keybind to shortcut clicking this button has been indicated to be a desired feature, due to the reliance on frequently clicking it.
- Scrolling within the webpage could be alleviated by removing scrolling of the full webpage (for the tool webpage). That would ensure there is always a 'check document' button at the bottom, while still allowing the flexibility of scrolling error messages or the code editor.
- More information should be provided by some error messages, with emphasis on those related to tables or tagging. By providing more information upfront, there will be less confusion about where to progress from a specific point.
- The error extracts being less helpful in the context of syntax errors should be alleviated by providing the line and character position information. In the future, if there is enough time, an automated fix feature could be implemented.
- Text sizes being too small can be solved by simply increasing their size. There is also potential for choosing different font families that fit the context better. Some consideration should be put towards whether the browser should be relied on for resizing of elements, rather than LinT_EX implementing it.
- The colour scheme was indicated to have not been distinct enough for dyslexia users, which can be alleviated by choosing and applying a more accessible colour scheme for this user group.

5.5 Hypotheses testing and discussion

It can be claimed that, as there is statistical significance in applying student's paired t-test (section 5.3.3), a right-tailed test can be used to ascertain that there is a definitive improvement in the screen reader accessibility of the documents after applying LinTeX. Additionally, based on section 5.3.2, it can be concluded that there was a significant improvement in the document's screen reader accessibility after applying LinTeX, as it showed previously difficult aspects becoming simpler to understand, with minimal difficult aspects in the accessible document. Consequently, we choose not to reject the hypothesis **H1**.

The hypothesis **H2** is a grey area, due to the magnitude of accessibility-related difficulties found in the Table 9. However, proposed solutions have been provided in areas of improvement for LinTeX usability (section 5.4.3), which indicate processes that may be undertaken to solve many of the identified concerns. As many of these solutions are resolvable with little effort, it suggests the accessibility concerns LinTeX faces are minor.

It can be reasonably chosen to not reject the hypothesis **H3** as the majority of error messages were greatly useful based on Table 5.4.3, with only a few points made to provide more information upfront to avoid the user being at a standstill for what action to perform at a given point and providing more information regarding the location the error had occurred.

The hypothesis **H4** is arguable, as the main key points were often recalled by the participants. However, as many finer details were significantly wrong, the overall memorability of the inaccessible document's content could be considered low. Consequently, this would permit not rejecting the hypothesis **H4**.

6 Discussion and reflection

- things that went wrong for user study
- things that could have been planned better
- GUI design

7 Conclusions

7.1 Project summary

7.2 Project limitations

In an ideal world, it would have been possible to have created a project that could target any and all $\text{\LaTeX}$ accessibility concerns. Unfortunately, that is outwith the scope of the project, which has been defined as follows:

- LinTeX is not designed to handle the majority of parsing. Instead, it focuses on a few key elements that are related to the accessibility concerns targetted in our context. Otherwise, it handles a small amount of verification regarding the structure of individual lines and the overall document structure.
- This project does not aim to provide automated fixes for errors encountered.
- LinTeX does not contain an in-built screen reader. While some considerations were put aside for how this could be accomplished, it is not feasible to create an in-built screen reader with the timeframe allocated for this project, due to the vast considerations needed. It might even be a fair statement to consider it a project on its own.
- LinTeX is not intended to find usage of outdated or incorrect accessibility, but rather to enhance a document's screen reader accessibility through additions to the document.
- LinTeX is not intended to debug $\text{\LaTeX}$ code, as a dedicated integrated development environment (IDE) and $\text{\TeX}$ engine should be used for this instead.
- LinTeX expects code to compile in $\text{Lua}\text{\LaTeX}$ before its changes have been applied, with other $\text{\TeX}$ engines being considered outdated and consequently not supported for this context.
- LinTeX places emphasis on accessibility features for screen reader users.
- It is unfeasible to account for all tagging options in our context. Therefore, a few key options have been selected based on what would be expected to have the most impact.

7.3 Achievements

7.4 Future work

LinTeX has the potential to grow in a variety of manners, such as targetting the limitations in project limitations (section 7.2).

An in-built screen reader for LinTeX is an interesting proposition, with it likely being a project by itself due to the lack of finer details requiring vast amounts of consideration and the difficulties with finding similar materials, seeing as most references for an in-built screen reader appear to point to screen readers as a browser extension. However, it may be possible to make LinTeX work with those extensions to provide the ideal experience for the end user.

Automated fixes, with the addition of being able to fix several identical errors at once, would be a great feature to include in LinTeX. There is potential for this feature to work alongside the error handling of LinTeX, yet this has unfortunately not been explored further at this time.

Additionally, highlighting why LinTeX has recommended a change would make the experience a lot more insightful and better encourage developers to be inclusive of screen reader users in the future.

Finally, utilising HTMX to make LinTeX more responsive to the end user's actions would allow for a more immersive experience for the user, while keeping the technologies required at a reasonable minimum requirements.

7.5 Reflection

7.6 Final thoughts

References

- Authors, The go-latex (2020). *latex*. English. <https://pkg.go.dev/codeberg.org/go-latex/latex>. URL: <https://pkg.go.dev/codeberg.org/go-latex/latex>.
- Authors, The Bootstrap (2011). *Bootstrap*. English. <https://getbootstrap.com/>. URL: <https://getbootstrap.com/>.
- Authors, The Go (2009a). *Exec*. English. <https://pkg.go.dev/os/exec>. URL: <https://pkg.go.dev/os/exec>.
- (2009b). *html/template*. English. <https://pkg.go.dev/html/template>. URL: <https://pkg.go.dev/html/template>.
- (2009c). *net/http*. English. <https://pkg.go.dev/net/http>. URL: <https://pkg.go.dev/net/http>.
- Authors, The Gorilla (2023). *gorilla/mux*. English. <https://pkg.go.dev/github.com/gorilla/mux>. URL: <https://pkg.go.dev/github.com/gorilla/mux>.
- Center, Delaware Government Information (2024). *What is User Research and Why Should it be a Part of Your Process?* English. URL: <https://gic.delaware.gov/what-is-user-research-and-why-should-it-be-a-part-of-your-process/>.
- Clifton, Andy (2019). *The Accessibility LaTeX package*. English. <https://github.com/AndyClifton/accessibility>. URL: <https://github.com/AndyClifton/accessibility>.
- Doctor, Nathan **and** Jake Hoffner (2012). *CodeWars*. English. <https://www.codewars.com/>. URL: <https://www.codewars.com/>.
- Exeter, University of (2025). *Test Statistics*. English. URL: <https://libguides.exeter.ac.uk/c.php?g=729256&p=5313318>.
- Inc., Bergside (2023). *Flowbite*. English. <https://flowbite.com/>. URL: <https://flowbite.com/>.
- Kpym (2020). *latex-fast-compile*. English. <https://pkg.go.dev/github.com/kpym/latex-fast-compile>. URL: <https://pkg.go.dev/github.com/kpym/latex-fast-compile>.
- Martínez-Almeida, Manuel (2014). *Gin Web Framework*. English. <https://pkg.go.dev/github.com/gin-gonic/gin>. URL: <https://pkg.go.dev/github.com/gin-gonic/gin>.
- Murrish, Robert (2012). *?LaTeX is More Powerful than you Think - Computing the Fibonacci Numbers and Turing Completeness?* English. **in** URL: https://www.overleaf.com/learn/latex/Articles/LaTeX_is_More_Powerful_than_you_Think_-_Computing_the_Fibonacci_Numbers_and_Turing_Completeness.
- National Library of Medicine, Network of the (2026a). *Qualitative Data*. English. URL: <https://www.nlm.gov/resources/data-glossary/qualitative-data>.
- (2026b). *Quantitative Data*. English. URL: <https://www.nlm.gov/resources/data-glossary/quantitative-data>.
- Ravisankar, Vivek **and** Harishankaran Karunanidhi (2011). *HackerRank*. English. <https://www.hackerrank.com/>. URL: <https://www.hackerrank.com/>.
- Software, Big Sky (2020). *HTMX*. English. <https://htmx.org/>. URL: <https://htmx.org/>.
- Sullivan, Gail M. (2013). *?Analyzing and Interpreting Data From Likert-Type Scales?* English. **in** URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3886444/>.

- Tailwind Labs, Inc. (2017). *Tailwind CSS*. English. <https://tailwindcss.com/>. URL: <https://tailwindcss.com/>.
- Tang, Winston (2015). *LeetCode*. English. <https://leetcode.com/>. URL: <https://leetcode.com/>.
- University, Ulster (2026). *What is accessibility and why is it important?* English. URL: <https://www.ulster.ac.uk/accessibility-standards/what>.
- W3C (2018). *Web Content Accessibility Guidelines (WCAG) 2.2*. <https://www.w3.org/TR/WCAG22/>. A W3C Recommendation. URL: <https://www.w3.org/TR/WCAG22/>.
- Wadhwa, Raoul R. **and** Raghavendra Marappa-Ganeshan (2023). *?T Test?* English. **in** URL: <https://www.ncbi.nlm.nih.gov/books/NBK553048/>.

A Appendix

A.1 Ethical approval



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Background

The purpose of the Departmental and University Ethics Committees is to ensure that any investigations (regardless of subject domain) carried out by staff, or students that use human beings as participants are known to conform to the standards set by the professional bodies. These standards are aimed at ensuring that the rights, safety, and wellbeing of the participants is taken into account at all times. The Departmental Ethics Committee members are Andrew Abel, Daniel Thomas, Emma Nicol, John Connaghan, Karen Renaud, Lisa McCann, Louise McCullagh, Marc Roper, Mark Dunlop, Martin Halvey, Morven Miller, Pejman Saeghe, Ryan Gibson, and Yingying Zhao. You may contact the committee using the following email address: cis-ethics@strath.ac.uk

What you need to do

It is important to realise that any investigation that involves human beings as participants - even something as apparently harmless as a questionnaire or simple user study being undertaken as part of a project or your research - has to be granted ethical approval. To gain this approval you need to complete the "[New application](#)" process at the end of this page and upload supporting documentation which addresses the following points:

- Title of research:
- Summary of research (short overview of the background and aims of this study):
- How will participants be recruited?
- What will the participants be told about the proposed research study?
Either upload or include a copy of the briefing notes issued to participants. In particular this should include details of yourself, the context of the study and an overview of the data that you plan to collect, your supervisor, and contact details for the Departmental Ethics Committee.
- How will consent be demonstrated?
Either upload or include here a copy of the consent form/instructions issued to participants. It is particularly important that you make the rights of the participants to freely withdraw from the study at any point (if they begin to feel stressed for example), nor feel under any pressure or obligation to complete the study, answer any particular question, or undertake any particular task. Their rights regarding associated data collected should also be made explicit.
- What will participants be expected to do?
Either upload or include a copy of the instructions issued to participants along with a copy of or link to the survey, interview script or task description you intend to carry out. Please also confirm (where appropriate) that your supervisor has seen and approved both your planned study and this associated ethics application.
- What data will be collected and how will it be captured and stored?
In particular indicate how adherence to the Data Protection Act and the General Data Protection Regulation (GDPR) - https://en.wikipedia.org/wiki/General_Data_Protection_Regulation - will be guaranteed and how participant confidentiality will be handled.
- How will the data be processed?
(e.g. analysed, reported, visualised, integrated with other data, etc.) Please pay particular attention to describing how personal or sensitive data will be handled and how GDPR regulations will be met.
- How and when will data be managed or disposed of?
Either upload a copy of your data management plan or describe how data will be disposed.

The form should be submitted well in advance of any planned study. Requests are normally processed within one working week, but may take longer than this for more complex studies or at certain times of the year.

The points above cover most studies carried out within the Department, but if you are in any doubt then you should read the [University's Code of Practice on Investigations on Human Beings](#) to make sure that there are not any other issues you should be considering. If you are uncertain about any aspect of your study then discuss this initially with your supervisor but also feel free to contact the Departmental Ethics Committee. More significant studies may need to be referred to the University Ethics Committee (UEC) and are likely to take much longer to process.

For the participant information sheets and consent form, you may find the pro-formas available from the UEC page useful <https://www.strath.ac.uk/ethics/> However, please take care to amend these appropriately and pay particular attention to the approval and contact details.

Contact Details

email: cis-ethics@strath.ac.uk

[New application](#)

ID	TITLE	STATUS	ACTION	RELATED IDS	LAST SUBMITTED	LAST REVIEWED
3188	LinTeX: Promoting Accessibility in LaTeX Documents	APPROVED	View Amend		2026-03-04 03:38:29	2026-03-09 12:49:45

A.2 Ethics application



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Application ID: 3188

Title of research:

LinTeX: Promoting Accessibility in LaTeX Documents

Summary of research (short overview of the background and aims of this study):

LinTeX is a new website being developed to combat accessibility problems in LaTeX, with a focus on screen readers. This study aims to understand its effectiveness at targeting this problem and highlighting solutions to its users. Additionally, it aims to identify areas for improvement.

How will participants be recruited?

Participants will be recruited via email and the 4th Year CompSci Strath 2025/26 Discord server in a non-invasive manner. The following message will be used in both contexts:

"Dear <Participant>,"

I would like your views on a tool I'm making that helps people write better LaTeX documents. In particular, it helps authors make documents that meet accessibility needs, and it helps authors with accessibility needs write their LaTeX documents.

LinTeX is a tool being developed to combat this dilemma, placing emphasis on accessibility for screen readers. This user study will determine its effectiveness in doing so.

This user study will involve an in-person think-aloud session taking place on the University of Strathclyde's campus. Further details can be found in the Participant Information Sheet (PIS) attached.

To participate in the user study, you should be a member of the University of Strathclyde, or another higher education facility. Some prior experience with LaTeX will be expected, and participants should be 18-years old or above.

Participation is entirely voluntary and will not affect you in any courses or how you are treated. If you could spare around 30 minutes to participate in the user study, it would be greatly appreciated.

If you are interested in participating in the user study, please contact me via email.

My contact details:

Mohammad Danyal
mohammad.danyal.2023@uni.strath.ac.uk

My supervisor's details:

Brian Mitchell
brian.mitchell@strath.ac.uk

Kind regards,

Mohammad Danyal"

- '<Participant>' will be replaced with the group/individual being contacted.

What will the participants be told about the proposed research study? Either upload or include a copy of the briefing notes issued to participants. In particular this should include details of yourself, the context of the study and an overview of the data that you plan to collect, your supervisor, and contact details for the Departmental Ethics Committee.

PDF File: [View document](#)

Details listed on Participant Information Sheet (PIS).

How will consent be demonstrated? Either upload or include here a copy of the consent form/instructions issued to participants. It is particularly important that you make the rights of the participants to freely withdraw from the study at any point (if they begin to feel stressed for example), nor feel under any pressure or obligation to complete the study, answer any particular question, or undertake any particular task. Their rights regarding associated data collected should also be made explicit.

PDF File: [View document](#)

Details listed on Consent Form.

What will participants be expected to do? Either upload or include a copy of the instructions issued to participants along with a copy of or link to the survey, interview script or task description you intend to carry out.

PDF File: [View document](#)

PDF File: None.

Details listed on the User Study Script.

What data will be collected and how will it be captured and stored? In particular indicate how adherence to the Data Protection Act and the General Data Protection Regulation (GDPR) will be guaranteed and how participant confidentiality will be handled.

Several pieces of data will be stored:

1. Assigning participants unique IDs. This section will have information such as their name and email address (to contact the participant for arranging a time/date). Participant IDs will be created when the participant has requested to join the user study.
2. Extracting details from the user study (such as answers to questions, audible points made by the participant, or unspoken points of note such as how long a specific task has taken to complete) and assigning it to the relevant participants' ID, consequently ensuring related data can be deleted as requested. All details for this group will be acquired from the user study.
3. Anonymising the data from the user study to be used in the final report. This section will have no participant IDs and will have no relation to any participant of the user study. Data will be anonymised at the earliest practicable stage.

Each group of data, as numbered above, will be stored only in separate folders on the university's OneDrive, within password-protected folders, ensuring a folder for each numbered section.

How will the data be processed? (e.g. analysed, reported, visualised, integrated with other data, etc.) Please pay particular attention to describing how personal or sensitive data will be handled and how GDPR regulations will be met.

Anonymised data will be inserted into the final project report after being thoroughly analysed.

Some examples may include:

- How often a listening, or a section of the listening, has been repeated at a stage;
- How often and where a section of the listening has been paused;
- Comparing answers with questions from the initial hearing to the hearing after having used the LinTeX tool;
- Analysing how well the overall document was understood initially to determine difficulties with screen readers;
- Analysing correlations between a user's experience with LaTeX and screen readers with the results from the other components of the user study;
- Mathematical analysis of Likert scale data including frequencies, median, percentages, and mode;
- Comparison of Likert scale data from before and after the think-aloud session.

Results will be presented to minimise the chance of any participant being identifiable. Additionally, any qualitative results will be anonymised.

How and when will data be disposed of? Either upload a copy of your data management plan or describe how data will be disposed.

PDF File: None.

Physical notes taken during the user study will be digitalised as soon as reasonably possible. Once digitalised, they will be securely destroyed.

Electronic data will be securely destroyed on 1st August 2026.

Supervisor's name (students need to complete/staff should probably put N/A).

Brian Mitchell

☒ I confirm that my supervisor has seen and approved both my planned study and this associated ethics application.

A.3 Participant information sheet

Participant Information Sheet for LinTeX User Study

[FOR USE WITH STANDARD PRIVACY NOTICE FOR RESEARCH PARTICIPANTS]

Name of department: Computer and Information Sciences

Title of the study: LinTeX: Promoting Accessibility in LaTeX Documents

Introduction

I am Mohammad Danyal, a final year Undergraduate Computer Science student at the University of Strathclyde. I can be contacted by email via mohammad.danyal.2023@uni.strath.ac.uk should any concerns or questions arise regarding the study, or if you wish to withdraw your participation.

What is the purpose of this research?

Modern LaTeX documents tend to use PDFLaTeX as opposed to LuaLaTeX. Thus, they often require workarounds to implement features and lack accessibility support.

LinTeX is a prototype to combat accessibility with emphasis placed on screen readers, and avoiding counter-productive workarounds.

This user study will identify the effectiveness of a new tool, LinTeX, to promote accessibility in LaTeX documents, as well as its usability to a wider audience.

Do you have to take part?

Participation in the research is entirely voluntary.

Refusing to participate in or withdrawing from the study will not affect the ways an individual is treated.

Participants may withdraw from the study at any stage without providing a reason.

If you wish to withdraw from the study at any stage, please contact mohammad.danyal.2023@uni.strath.ac.uk.

What will you do in the project?

Participants will be asked to meet in-person in the University of Strathclyde's campus at a pre-arranged time and date.

Participants will be provided with a LaTeX document at the start of the user study. Participants will then listen to a screen reader's output for the LaTeX document.

Afterwards, participants will be asked questions about their thoughts on the output.

Participants will then be requested to use LinTeX in a think-aloud session, where you say what you are thinking, to improve the document's accessibility through improvements suggested by LinTeX.

Participants will finally repeat the initial step of listening to a screen reader's output for the improved document and answer questions regarding their opinion of what went well and what could have been improved.

Why have you been invited to take part?

Participants of the research study may benefit from the LinTeX tool and have been asked to assist in its development through an in-person think-aloud user study.

Participants must be a member of the University of Strathclyde, or another higher education facility. Travel costs cannot be reimbursed.

Participants should be 18-years old or above.

Participants will require prior experience with LaTeX to participate in the study and have been selected based on these criteria.

What information is being collected in the project?

Personal information that may be stored include the participants' names and email addresses. This datum is stored to ensure communication is thorough in arranging a time and date to participate in the study. Additionally, some information stored about the participants' usage of the system or answers to questions may be personal.

Non-identifiable information that may be stored include anonymised information extracted from the user study.

No audio or video recording will be created. No transcripts will be created.

Who will have access to the information?

Only the researcher will have access to any Personally Identifiable Information (PIP). Anonymised information may be included in the research paper and shared with the supervisor.

Where will the information be stored and how long will it be kept for?

All data will be stored only in password-protected folders on OneDrive, with Personally Identifiable Information (PIP) and anonymous data stored in separate folders.

All data stored in these folders will be deleted on 1st August 2026. However, any data that is anonymised and included in the study may not be possible to remove.

Physical notes may be taken during the user study, which will be digitalised within a week. Once digitalised, it will be securely destroyed.

Thank you for reading this information – please ask any questions if you are unsure about what is written here.

All personal data will be processed in accordance with applicable Scottish data protection legislation. Please read our [Privacy Notice for Research Participants](#) for more information about your rights under the legislation.

What happens next?

If you are interested in the project, or continuing with the user study, please contact me by email via mohammad.danyal.2023@uni.strath.ac.uk. Participants will be requested to sign a consent form prior to participation in the user study.

All information collected in the user study will be anonymised before being published.

If participants are interested in participating in any future related studies, the aforementioned email address can be used to register your interest.

If you are no longer interested in participating in the project, no further steps will be required. Thank you for your time.

Researcher contact details:

Mohammad Danyal

mohammad.danyal.2023@uni.strath.ac.uk

Chief Investigator details:

Brian Mitchell

brian.mitchell@strath.ac.uk

This research was granted ethical approval by the Department of Computer and Information Sciences Ethics Committee.

If you have any questions/concerns, during or after the research, or wish to contact an independent person to whom any questions may be directed or further information may be sought from, please contact: cis-ethics@strath.ac.uk

A.4 Consent form

Consent Form for LinTeX User Study

Name of department: Computer and Information Sciences

Title of the study: LinTeX: Promoting Accessibility in LaTeX Documents

Please check each item to indicate agreement.

- ☐ I confirm that I have read and understood the Participant Information Sheet for the above project and the researcher has answered any queries to my satisfaction.
- ☐ I confirm that I have read and understood the [Privacy Notice for Participants in Research Projects](#) and understand how my personal information will be used and what will happen to it (i.e. how it will be stored and for how long).
- ☐ I understand that my participation is voluntary and that I am free to withdraw from the project at any time, up to the point of completion, without having to give a reason and without any consequences.
- ☐ I understand that I can request the withdrawal from the study of some personal information and that whenever possible researchers will comply with my request. This includes the following personal data:
 - ☐ My personal information collected from the think-aloud session.
 - ☐ My personal information from answers to questions.
 - ☐ My personal information from transcripts.
- ☐ I understand that anonymised data (i.e. data that do not identify me personally) cannot be withdrawn once they have been included in the study.
- ☐ I understand that any information recorded in the research will remain confidential and no information that identifies me will be made publicly available.
- ☐ I consent to being a participant in the project.

(PRINT NAME)	
Signature of Participant:	Date:

A.5 User study script

Script for LinTeX User Study

Name of department: Computer and Information Sciences

Title of the study: LinTeX: Promoting Accessibility in LaTeX Documents

Introduction

Hello, I am Mohammad Danyal, a final year Computer Science student at the University of Strathclyde. I am the Researcher for this user study.

As a general overview of what this user study will consist of:

- Introductory questions to gauge experience levels
- An initial phase for listening to the document through a screen reader.
- Answering questions about how easy or difficult the document is to understand.
- Utilising the proposed tool, LinTeX, to make the document more accessible to screen reader users.
- A final listening of the document, with the changes requested by the tool having been applied.
- Answering further questions about whether the tool had made an impact on the difficulty for understanding the document as a screen reader user.

These tasks will be explained in more detail as we approach them.

If you have any questions as we go through the process, feel free to chime in.

You have the right to opt to not answer questions as you wish, as stated in the consent form.

Introductory questions

- On a scale of 1 to 5, how often do you use LaTeX?
 1. Never
 2. Rarely
 3. Sometimes
 4. Often
 5. Almost always
- On a scale of 1 to 5, how often do you use screen readers?
 1. Never
 2. Rarely
 3. Sometimes
 4. Often
 5. Almost always
- On a scale of 1 to 5, how familiar are you with listening to audio recordings of speaking?
 1. Very unfamiliar
 2. Unfamiliar
 3. Neither unfamiliar/familiar
 4. Familiar
 5. Very familiar

Initial listening

For this step, I will have you listen to the output of the document before any changes have been made to it. If you have any device for listening to the output that you'd prefer to use, just make that preference known and it can be setup.

While listening to the output, if you would like to make any points of interest, simply indicate that and the audio output can be paused. Additionally, if you would like to repeat any sections or the full recording, feel free to ask.

When the recording has been complete, and no requests are made to repeat a section, we will move onto answering a few questions.

Questions [initial listening]

- Provide a general overview of what the document discussed.
- What parts of the listening were simple to understand? Why?
- What parts of the listening were difficult to understand? Why?
- On a scale of 1 to 5, how would you rate the quality of the audio output? (i.e. in terms of the content of its output)
 1. Very poor
 2. Poor
 3. Neither poor/good
 4. Good
 5. Very good
- How well could you understand the purpose of tables, images or figures in the listening?

Using LinTeX

In this step, we will perform a think aloud session. In case you have not participated in one before, here is a general summary:

- You will be tasked to perform some action.
- You should audibly state your thought process as you go through the task. For example, this may be in the form of the stating having trouble with something.
- There will be minimal intervention from me.

The task is in 3 parts:

1. Upload the document to LinTeX by navigating its GUI
2. Improve the document's accessibility by following what LinTeX provides.
3. Export the document.

You will be provided with the file's location, as well as the LinTeX tool to work with.

Final listening

Similar to before, you will be tasked with listening to the output of the document. This time, after changes have been applied through the LinTeX tool.

If you would like to make any points of interest, simply indicate that and the audio output can be paused. Additionally, if you would like to repeat any sections or the full recording, feel free to ask.

When the recording has been complete, and no requests are made to repeat a section, we will move on to answering the final questions.

Questions [final listening]

- What parts of the listening were simple to understand? Why?
- What parts of the listening were difficult to understand? Why?

- On a scale of 1 to 5, how would you rate the quality of the audio output? (i.e. in terms of the content of its output)

1. Very poor
2. Poor
3. Neither poor/good
4. Good
5. Very good

- How well could you understand the purpose of tables, images or figures in the listening?

- On a scale of 1 to 5, how likely are you accommodate for screen reader users in the future?

1. Very unlikely
2. Unlikely
3. Neither unlikely/likely
4. Likely
5. Very likely

Exit

Thank you for your time. You have been of great help for this user study. If you wish to see where the project may end up or have any questions, my email is on the Participant Information Sheet under the Researcher.

A.6 Inaccessible document

```

\documentclass[draft]{article}
\usepackage{graphicx}
\begin{document}
\section{Introduction}
This document is intended to be read by a screen reader to indicate related
accessibility concerns for LaTeX users.

\section{Image Examples}
We will have 3 examples for our context:
\begin{itemize}
  \item An example of an image that should contain alternative text ('alt
text'). Alternative text is a descriptive text used in place of an image or
graphic.
  \item An example of an image that has actual text within it that must be
read
  \item An example of an image that is purely for decorative purposes i.e.
it should not be read.
\end{itemize}
These will be in no particular order. Some filler text will be included to make
them appear more natural.

```

Every day for a year, I have practiced my speech. I have used many techniques to improve my speaking skills, such as socialising, taking lessons, trying out tongue twisters, amongst many other things.\\

```

\includegraphics{tongue_twister.jpg}\\ % use actual text
That was an example of a tongue twister I have mastered.

```

This house is for sale. Its market price is currently \\$100,000, but it is estimated to rise to \$250,000 in a year.\\

```

\includegraphics{simple_house.png}\\ % use alt text

```

I have many pets, but my favourite is my cat. Cats are great!\\

```

\includegraphics{cat.png}\\ % use decorative (artifact) or alt text

```

```

\section{Table Examples}
We will have 2 examples for our context:
\begin{itemize}
  \item An example of a table that should have its heading highlighted.
  \item An example of a table that is purely for decorative purposes i.e. it
should not be read.
\end{itemize}
These will be in no particular order. Some filler text will be included to make
them appear more natural.

```

Names are interesting. How much more common is the name Noah compared to Muhammad in Scotland?\\

```

\begin{tabular}{ccc} % spoken table
  Year & Muhammad Position & Noah Position \\ \hline
  2024 & 2nd & 1st \\
  2023 & 11th & 2nd \\
  2022 & 23rd & 1st \\
  2021 & 43rd & 2nd \\
  2020 & 31st & 2nd \\
\end{tabular}\\

```

From this table, we can see that originally Muhammad was a far less common name than Noah, but in the period of 5 years Muhammad has drastically rose from 31st place to 2nd place, making it a strong competitor against Noah which has retained a top 2 placing across all years.

Here are my grades from this year, compared to my friends' grades.\\

```

\begin{tabular}{ccccc} % presentation table
  Name & CS913 & ML123 & MS019 & HI129 \\ \hline
  James & 12\% & 20\% & 80\% & 67\% \\
  Patricia & 80\% & 90\% & 31\% & 19\% \\
\end{tabular}

```

```
John & 34\% & 89\% & 55\% & 98\%
\end{tabular}\\
I think John did the best out of them!
\end{document}
```

A.7 Accessible document

```

\DocumentMetadata{
  tagging = on,
  tagging-setup = {activate/all},
  pdfstandard = ua-2,
  lang = en-GB,
  pdfversion = 2.0
}
\documentclass[draft]{article}
\usepackage{graphicx}
\begin{document}
\section{Introduction}
This document is intended to be read by a screen reader to indicate related
accessibility concerns for LaTeX users.

\section{Image Examples}
We will have 3 examples for our context:
\begin{itemize}
  \item An example of an image that should contain alternative text ('alt
text'). Alternative text is a descriptive text used in place of an image or
graphic.
  \item An example of an image that has actual text within it that must be
read
  \item An example of an image that is purely for decorative purposes i.e.
it should not be read.
\end{itemize}
These will be in no particular order. Some filler text will be included to make
them appear more natural.

```

Every day for a year, I have practiced my speech. I have used many techniques to improve my speaking skills, such as socialising, taking lessons, trying out tongue twisters, amongst many other things.\\

```

\includegraphics[actualtext={I slit the sheet, the sheet I slit, and on the
slitted sheet I sit}]{tongue_twister.jpg}\\ % use actual text

```

That was an example of a tongue twister I have mastered.

This house is for sale. Its market price is currently \\$100,000, but it is estimated to rise to \$250,000 in a year.\\

```

\includegraphics[alt={An image of the house being sold.}]{simple_house.png}\\ %
use alt text

```

I have many pets, but my favourite is my cat. Cats are great!\\

```

\includegraphics[artifact]{cat.png}\\ % use decorative (artifact) or alt text

```

```

\section{Table Examples}
We will have 2 examples for our context:
\begin{itemize}
  \item An example of a table that should have its heading highlighted.
  \item An example of a table that is purely for decorative purposes i.e. it
should not be read.
\end{itemize}
These will be in no particular order. Some filler text will be included to make
them appear more natural.

```

Names are interesting. How much more common is the name Noah compared to Muhammad in Scotland?\\

```

\tagpdfsetup{table/header-rows={1}}
\begin{tabular}{ccc} % spoken table
  Year & Muhammad Position & Noah Position \\ \hline
  2024 & 2nd & 1st \\
  2023 & 11th & 2nd \\
  2022 & 23rd & 1st \\
  2021 & 43rd & 2nd \\
  2020 & 31st & 2nd \\

```

\end{tabular}\\\

From this table, we can see that originally Muhammad was a far less common name than Noah, but in the period of 5 years Muhammad has drastically rose from 31st place to 2nd place, making it a strong competitor against Noah which has retained a top 2 placing across all years.

Here are my grades from this year, compared to my friends' grades.\\

\tagpdfsetup{table/tagging=presentation}

\begin{tabular}{ccccc} % presentation table

 Name & CS913 & ML123 & MS019 & HI129 \\ \hline

 James & 12\% & 20\% & 80\% & 67\% \\

 Patricia & 80\% & 90\% & 31\% & 19\% \\

 John & 34\% & 89\% & 55\% & 98\%

\end{tabular}\\\

I think John did the best out of them!

\end{document}

A.8 PEG script

```

{
    package main

    var groupStack Stack[Component] = newStack[Component]()
    var documentBuilder IDocumentBuilder = newDocumentBuilder()
    var argumentStack Stack[string] = newStack[string]()
}

/**
 * This section deals with the overall formatting of the document.
 * This includes enforcing an expected format is used, and handling of every
component of the document.
 * Additionally, it is necessary to return the overall Document struct created
to represent the parsed document.
 */
// Matches against overall document structure
Input <- &{
    // re-initialise data
    groupStack = newStack[Component]()
    documentBuilder = newDocumentBuilder()
    argumentStack = newStack[string]()
    return true, nil
} text prerequisite:PrerequisiteContent text docclass:DocumentClass text
preamble:Preamble text content:DocumentContent text EOF {
    // creating coordinates for potential error handling
    startCoord, endCoord, err := GetCoordinates(c.pos.line, c.pos.col, c.text)
    if err != nil {
        return nil, DummyError(err)
    }

    // handling prerequisite
    if prerequisite != nil {
        for _, component := range prerequisite.([]Component) {
            errFn := documentBuilder.addPrerequisiteLine(component)
            if errFn != nil {
                return nil, errFn(*startCoord, *endCoord)
            }
        }
    }

    // handling document class
    if docclass != nil {
        // add to the document
        errFn := documentBuilder.addDocumentClass(*docclass.(*Line))
        if errFn != nil {
            return nil, errFn(*startCoord, *endCoord)
        }
    }

    // handling preamble
    if preamble != nil {
        for _, line := range preamble.([]Line) {
            var component Component = &line
            errFn := documentBuilder.addPreambleLine(component)
            if errFn != nil {
                return nil, errFn(*startCoord, *endCoord)
            }
        }
    }

    // handling document content
    if content != nil {
        for i, component := range content.([]Component) {
            var errFn CreateError

```

```

        if i + 1 < len(content.([]Component)) {
            errFn = documentBuilder.addPreambleLine(component)
        } else {
            errFn = documentBuilder.addDocumentContent(component)
        }
        if errFn != nil {
            return nil, errFn(*startCoord, *endCoord)
        }
    }
}

// additional checks over the document occur during creation
document, errFn := documentBuilder.buildDocument()
if errFn != nil {
    return nil, errFn(*startCoord, *endCoord)
}
return document, nil
}

// Handles commands that precede the document class
// These should be rarely used in most LaTeX documents
PrerequisiteContent <- lines:MultiLaTeXLines {
    return lines, nil
}

// Handles the document class i.e. overall categorisation of the LaTeX document
DocumentClass <- command_start name:`documentclass` args:MultiArgument {
    startCoord, endCoord, err := GetCoordinates(c.pos.line, c.pos.col, c.text)
    if err != nil {
        return nil, DummyError(err)
    }

    // parse data
    arguments := args.([]Argument)
    line := newLine(string(name.([]byte)), arguments, *startCoord, *endCoord)
    return line, nil
} / text {
    return nil, nil
}

// Handles metadata (i.e. the head) about the LaTeX document
Preamble <- lines:MultiSimpleLaTeXLines {
    return lines, nil
}

// Handles the overall content (i.e. the body) of the LaTeX document
DocumentContent <- lines:MultiLaTeXLines {
    return lines, nil
}

// Ensures the document does not match any further
// i.e. it ensures the End-of-File has been reached
EOF <- !.

/**
 * This section deals with LaTeX lines that require grouped contexts.
 * For example, all document content should be grouped under the 'document'.
 */
MultiLaTeXLines <- lines:LaTeXLine* {
    // creating coordinates for potential error handling
    startCoord, endCoord, err := GetCoordinates(c.pos.line, c.pos.col, c.text)
    if err != nil {
        return nil, DummyError(err)
    }
}

```

```

var result []Component
stack := newStack[Component]()
for _, line := range lines.([]any) {
    if line == nil {
        continue
    }

    // Handling ending groups
    coordinate, ok := line.(Coordinate)
    if ok {
        group, err := stack.Pop()
        if err != nil {
            return nil, err(*startCoord, *endCoord)
        }

        // updating group's end
        group.SetEndCoordinate(coordinate)
        continue
    }

    // adding component to pre-existing groups if necessary
    component, ok := line.(Component)
    if !ok || component == nil {
        return nil,
SERVER_RESPONSIBLE_NON_COMPONENT_DATA_STRUCTURE(*startCoord, *endCoord)
    }
    if !stack.IsEmpty() {
        peek, err := stack.Peek()
        if err != nil {
            return nil, err(*startCoord, *endCoord)
        }
        peer, ok := peek.(*Group)
        if !ok {
            return nil,
SERVER_RESPONSIBLE_STACK_CONTAINS_NON_GROUP_ELEMENT(*startCoord, *endCoord)
        }
        peer.AddComponent(component)
        component.SetGroup(peer)

        // add to the result if this is the topmost layer
    } else {
        result = append(result, component)
    }

    // handling grouped contexts
    _, ok = component.(*Group)
    if ok {
        stack.Push(component)
        continue
    }
}
return result, nil
}

LaTeXLine <- text line:(BeginGroup / EndGroup / SimpleLaTeXLine) text {
    return line, nil
}

BeginGroup <- command_start `begin` args:MultiArgument {
    // retrieving useful data
    startCoord, endCoord, err := GetCoordinates(c.pos.line, c.pos.col, c.text)
    if err != nil {
        return nil, DummyError(err)
    }
}

```

```

}
arguments := args.([]Argument)

// retrieving name of the group
var name string
found := false
for i, arg := range arguments {
    class, ok := arg.(*ClassArgument)
    if ok {
        name = strings.Trim(class.GetValue().(string), WHITESPACE)
        arguments[i] = nil
        found = true
        break
    }
}
if !found {
    return nil, MISSING_BEGIN_GROUP_NAME(*startCoord, *endCoord)
}

// handling groups ordering
arguments = RemoveNilElements[Argument](arguments, nil)
group := newGroup(name, arguments, *startCoord, *endCoord)
groupStack.Push(group)
return group, nil
}

EndGroup <- command_start `end` args:MultiArgument {
    // creating coordinates for potential error handling
    startCoord, endCoord, err := GetCoordinates(c.pos.line, c.pos.col, c.text)
    if err != nil {
        return nil, DummyError(err)
    }
    arguments := args.([]Argument)

    // retrieving name of the group
    var name string
    found := false
    for _, arg := range arguments {
        class, ok := arg.(*ClassArgument)
        if ok {
            name = strings.Trim(class.GetValue().(string), WHITESPACE)
            found = true
            break
        }
    }
    if !found {
        return nil, MISSING_END_GROUP_NAME(*startCoord, *endCoord)
    }

    // checking a begin statement exists
    peek, err := groupStack.Peek()
    if err != nil {
        return nil, GROUP_BEGIN_MISSING(*startCoord, *endCoord)
    }

    // checking the group beginning matches its end
    if peek.GetName() != name {
        return nil, GROUP_NAME_MISMATCH(*startCoord, *endCoord)
    }
    groupStack.Pop()

    // requires a non-nil value to indicate it is not an error
    // we supply endCoord so we can update the group coordinates to account for
    all the inner lines

```

```

    return *endCoord, nil
}

/**
 * This section handles simple LaTeX lines that do not require knowing about the
 * other lines to be processed.
 */
MultiSimpleLaTeXLines <- lines:SimpleWrappedLaTeXLine* {
    return AnyInterfaceToTSlice[Line](lines), nil
}

SimpleWrappedLaTeXLine <- text line:SimpleLaTeXLine text {
    return *line.(*Line), nil
}

// Handles any LaTeX line and provides it a data structure to be parsed further
// disallow 'documentclass', 'begin', and 'end' since they are handled
// separately
SimpleLaTeXLine <- command_start !`documentclass` !`begin` !`end` name:word+
args:MultiArgument {
    startCoord, endCoord, err := GetCoordinates(c.pos.line, c.pos.col, c.text)
    if err != nil {
        return nil, DummyError(err)
    }

    arguments := args.([]Argument)
    line := newLine(AnyInterfaceToString(name), arguments, *startCoord,
*endCoord)
    return line, nil
}

/**
 * This section deals with argument handling of individual LaTeX commands.
 * Namely, components within them that are surrounded by curly braces '{}' or
 * square brackets '[]'.
 * This section also includes handling for multiple arguments one after the
 * other.
 */

// Handles zero or more arguments in a row, directly one after the other
MultiArgument <- arguments:Argument* {
    return AnyInterfaceToTSlice[Argument](RemoveNilElements[any](arguments.
([]any), nil)), nil
}

// Handles either possible arguments, ensuring a uniform method of management
Argument <- argument:(OptionArgument / ClassArgument) {
    argumentStack = newStack[string]()
    return argument, nil
}

// For our purposes, a class argument is defined as one surrounded by curly
// braces '{}'
ClassArgument <- '{' _ class:ArgumentContent _ '}' {
    startCoord, endCoord, err := GetCoordinates(c.pos.line, c.pos.col, c.text)
    if err != nil {
        return nil, DummyError(err)
    }
    if !argumentStack.IsEmpty() {
        return nil, INNER_ARGUMENT_END_MISSING(*startCoord, *endCoord) // custom
err later
    }
    argument := newClassArgument(class.(string))
    return argument, nil
}

```

```

// Handling if the closing brace '}' is missing
} / '{' _ ArgumentContent {
  startCoord, endCoord, err := GetCoordinates(c.pos.line, c.pos.col, c.text)
  if err != nil {
    return nil, DummyError(err)
  }
  return nil, CLASS_ARGUMENT_END_MISSING(*startCoord, *endCoord)
} / ArgumentContent _ '}' {
  startCoord, endCoord, err := GetCoordinates(c.pos.line, c.pos.col, c.text)
  if err != nil {
    return nil, DummyError(err)
  }
  return nil, CLASS_ARGUMENT_START_MISSING(*startCoord, *endCoord)
}

// For our purposes, an option argument is defined as one surrounded by square
brackets '['']
OptionArgument <- '[' _ options:ArgumentContent _ ']' {
  argument := newOptionArgument(options.(string))
  return argument, nil
}

// Handling if the closing bracket ']' is missing
} / '[' _ ArgumentContent {
  startCoord, endCoord, err := GetCoordinates(c.pos.line, c.pos.col, c.text)
  if err != nil {
    return nil, DummyError(err)
  }
  return nil, OPTIONS_ARGUMENT_END_MISSING(*startCoord, *endCoord)
} / ArgumentContent _ ']' {
  startCoord, endCoord, err := GetCoordinates(c.pos.line, c.pos.col, c.text)
  if err != nil {
    return nil, DummyError(err)
  }
  return nil, OPTIONS_ARGUMENT_START_MISSING(*startCoord, *endCoord)
}

/**
 * This section handles inner content of arguments. We do not need to parse the
 * data into their own structures, only match against required formatting.
 */

// Handling of the full inner content of an argument
ArgumentContent <- ArgumentPart* {
  return string(c.text), nil
}

// Handling of argument 'edges' such as {} or [], as well as allowing simple
LaTeX lines and ordinary text to pass through
ArgumentPart <- SimpleLaTeXLine / non_argument_content / ArgumentOpen /
ArgumentClose

// Handles opening argument edges, utilising a stack
ArgumentOpen <- str:argument_opener {
  chr := string(str.([]byte))
  argumentStack.Push(chr)
  return chr, nil
}

ArgumentClose <- &{
  // only match against the ending if there is a starting component
  // otherwise, we assume it is to be matched by the argument itself rather
  than its inner contents
  if argumentStack.IsEmpty() {

```

```

        return false, nil
    }
    return true, nil
} str:argument_closer {
    startCoord, endCoord, err := GetCoordinates(c.pos.line, c.pos.col, c.text)
    if err != nil {
        return nil, DummyError(err)
    }

    argMapping := map[string]string{
        "[": "]",
        "{": "}",
    }

    closeChr := string(str.([]byte))
    openChr, errFn := argumentStack.Pop()
    if errFn != nil {
        return nil, errFn(*startCoord, *endCoord)
    }
    expectedCloseChr, ok := argMapping[openChr]
    if !ok {
        return nil, INNER_ARGUMENT_END_MISSING(*startCoord, *endCoord)
    }
    if expectedCloseChr != closeChr {
        return nil, INNER_ARGUMENT_MISMATCH(*startCoord, *endCoord)
    }
    return nil, nil
}

/**
 * This section deals with character classes which are used to indicate a group
of characters, or the lack thereof.
 * It serves a purpose of improving maintainability and readability of the
overall PEG script.
 */
// Handles argument edges
argument_opener <- [{"["]
argument_closer <- [{"}"]
non_argument_content <- [{"^\\{\\}"}]

// Handles supported commands, currently only including \
command_start <- [{"\\"]
non_special <- [{"^\\"]
text "zero or more non-special characters" <- (command_start (& non_whitespace
non_word) / non_special)*

// Handles word characters
word <- [{"a-zA-Z0-9_"]
non_word <- [{"^a-zA-Z0-9_"]

// whitespace handling
non_whitespace <- [{"^ \\n\\t\\r"]
whitespace <- [{" \\n\\t\\r"]
_ "zero or more whitespace characters" <- whitespace*

```